Monosaccharides: Monosaccharides are carbohydrates that cannot be hydrolysed further and are also called simple sugars. Monosaccharides have general formula $C_n(H_2O)_n$. While there are many monosaccharides known only about 20 of them occur in nature. Some common examples are glucose, fructose, ribose, erythrose

Monosaccharides are further classified based on the functional group present (aldoses or ketoses) and the number of carbon present in the chain (trioses, tetroses, pentoses, hexoses etc...). If the carbonyl group is an aldehyde, the sugar is an aldose. If the carbonyl group is a ketone, the sugar is a ketose. The most common monosaccharides have three to eight carbon atoms.

Table 14.1 Different types of monosaccharides:

No. of carbon atoms in the chain	Functional group present	Type of sugar	Example
3	Aldehyde	Aldotriose	Glyceraldehyde
3	Ketone	Ketotriose	Dihydroxy acetone
4	Aldehyde	Aldotetrose	Erythrose
4	Ketone	Ketotetrose	Erythrulose
5	Aldehyde	Aldopentose	Ribose
5	Ketone	Ketopentose	Ribulose
6	Aldehyde	Aldohexose	Glucose
6	Ketone	Ketohexose	Fructose

14.1.3 Glucose

Glucose is a simple sugar which serves as a major energy source for us. It is the most important and most abundant sugar. It is present in honey, sweet fruits such as grapes and mangoes etc... Human blood contains about 100 mg/dL of glucose, hence it is also known as blood sugar. In the combined form it is present in sucrose, starch, cellulose etc.,

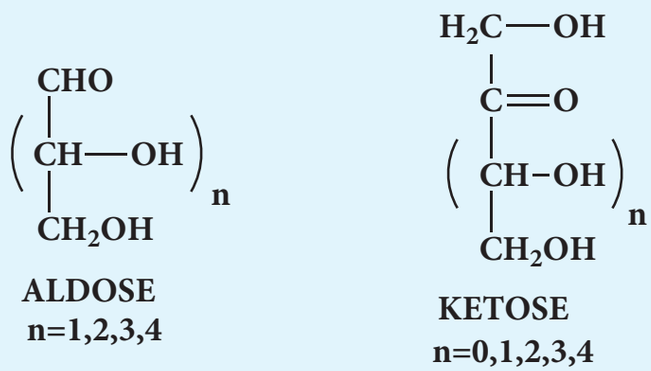


Figure 14.3 Structure of aldoses and ketoses

Preparation of glucose

1. When sucrose (*cane sugar*) is boiled with dilute H_2SO_4 in alcoholic solution, it undergoes hydrolysis and give glucose and fructose.

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

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